

KELSEY DAM, MB, CanadaWMO: **715720**Lat: **56.038N**Lon: **96.510W**Elev: **183**StdP: **99.15**Time Zone: **-6.00 (NAC)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.8	(c) -35.2	(d) -41.6	(e) 0.1	(f) -37.6	(g) -38.9	(h) 0.1	(i) -35.2	(j) 8.0	(k) -19.7	(l) 7.2	(m) -20.8	(n) 1.5	(o) 250	(p) 0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	27.3	17.0	25.7	16.4	24.0	15.7	19.1	24.5	17.9	23.1	16.9	22.1	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.6	21.0	15.9	11.5	19.9	14.7	10.7	18.9	55.1	24.4	51.2	22.9	48.0	22.2	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	7.0	6.3	-41.6	31.2	2.7	2.1	-43.5	32.7	-45.1	33.9	-46.6	35.1	-48.5	36.6
			-41.6	21.0	2.5	1.4	-43.4	22.0	-44.9	22.8	-46.3	23.6	-48.2	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.6	-22.2	-21.1	-13.2	-3.0	5.3	12.6	16.7	14.4	8.9	0.9	-10.7	-20.6
	DBStd	15.21	7.33	6.62	7.80	7.51	5.48	4.46	3.48	3.63	4.72	4.59	7.28	8.08
	HDD10.0	5102	999	871	718	391	165	23	1	6	76	283	620	949
	HDD18.3	7665	1258	1104	977	639	404	179	74	129	284	540	870	1207
	CDD10.0	517	0	0	0	1	20	100	207	143	44	2	0	0
	CDD18.3	38	0	0	0	0	1	6	22	8	2	0	0	0
	CDH23.3	496	0	0	0	0	16	116	252	96	17	0	0	0
	CDH26.7	79	0	0	0	0	3	17	44	12	2	0	0	0
Wind	WSAvg	3.0	2.7	2.8	3.0	3.3	3.6	3.3	2.9	2.9	3.0	3.2	2.9	2.6
Precipitation	PrecAvg	490	19	15	21	22	40	62	81	78	59	47	27	19
	PrecMax	684	38	34	59	52	82	104	136	199	98	104	66	33
	PrecMin	376	8	2	4	7	12	16	43	33	29	21	15	7
	PrecStd	66	7	7	13	11	16	19	28	38	22	21	11	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	-1.2	8.5	18.9	25.4	28.6	30.3	28.2	25.5	16.6	6.2	-0.6
		MCWB	-0.9	-2.7	4.1	9.9	14.0	17.0	19.4	18.5	17.6	10.4	3.7	-1.1
	2%	DB	-5.1	-4.0	5.3	13.9	21.7	26.3	27.7	26.0	21.9	12.7	2.9	-3.9
		MCWB	-5.6	-5.0	1.7	7.2	11.7	15.2	17.5	17.2	15.7	8.7	1.5	-4.4
	5%	DB	-7.6	-7.5	2.2	10.4	18.5	24.5	26.2	23.9	18.7	9.8	1.0	-6.3
		MCWB	-7.9	-8.3	-0.3	4.8	10.5	14.8	17.0	16.3	13.7	7.1	0.2	-6.7
	10%	DB	-10.5	-10.2	-0.4	7.9	15.2	22.1	24.5	21.6	16.4	7.3	-0.7	-8.4
		MCWB	-10.8	-10.9	-1.9	3.3	9.0	13.8	16.3	15.4	12.6	5.4	-1.3	-8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	-2.4	4.3	10.4	14.9	20.0	21.4	19.9	18.6	11.2	3.8	-1.1
		MCDB	-0.6	-1.4	8.0	18.3	22.9	24.3	27.5	25.2	24.0	14.5	5.5	-0.7
	2%	WB	-5.5	-5.0	2.0	7.7	12.8	17.2	19.3	18.6	16.2	9.2	1.7	-4.3
		MCDB	-5.1	-4.1	4.9	13.3	19.6	23.6	24.9	23.5	20.4	12.9	2.7	-3.9
	5%	WB	-7.9	-8.3	0.1	5.4	11.0	15.8	18.2	17.4	14.6	7.1	0.2	-6.6
		MCDB	-7.6	-7.6	1.8	9.7	17.2	22.1	23.5	22.2	18.1	9.5	0.8	-6.2
	10%	WB	-10.8	-10.9	-2.1	3.6	9.4	14.5	17.2	16.2	13.0	5.5	-1.3	-8.7
		MCDB	-10.5	-10.2	-0.3	7.5	14.9	20.6	22.3	20.1	16.2	7.3	-0.7	-8.5
Mean Daily Temperature Range	5% DB	MDBR	9.0	12.0	14.1	13.3	12.1	12.7	11.8	10.3	9.1	6.6	7.7	8.1
		MCDBR	8.9	12.7	14.9	15.2	17.4	16.8	15.0	14.2	12.5	11.4	8.2	7.3
		MCWBR	8.8	11.7	11.3	9.4	9.3	7.8	6.8	7.1	7.3	7.6	7.1	7.0
	5% WB	MCDBR	8.7	12.4	12.9	14.1	15.5	13.6	12.4	11.5	11.0	10.7	7.8	7.2
		MCWBR	8.6	11.5	10.0	9.0	8.8	7.4	6.4	6.7	7.0	7.4	6.8	6.9
Clear-Sky Solar Irradiance	taub		0.241	0.239	0.272	0.330	0.336	0.357	0.402	0.356	0.324	0.287	0.251	0.215
	taud		2.179	2.309	2.276	2.278	2.439	2.403	2.329	2.464	2.511	2.564	2.315	2.178
	Ebn at Noon		679	859	902	889	906	886	833	856	841	800	674	626
	Edh at Noon		55	72	97	114	103	109	115	95	78	58	48	42
All-Sky Solar Radiation	RadAvg		0.81	1.90	3.50	5.15	5.59	5.79	5.35	4.27	2.87	1.47	0.69	0.51
	RadStd		0.07	0.14	0.27	0.42	0.46	0.51	0.35	0.37	0.33	0.21	0.07	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page